

$$\begin{array}{ccc}
 f & \xrightarrow{H} & Hf \\
 \downarrow T_\tau & & \downarrow T_\tau \\
 T_\tau f & \xrightarrow{H} & H(T_\tau f) = T_\tau(Hf)
 \end{array}$$

$$T_\tau f(x) = f(x - \tau), \quad HT_\tau = T_\tau H$$